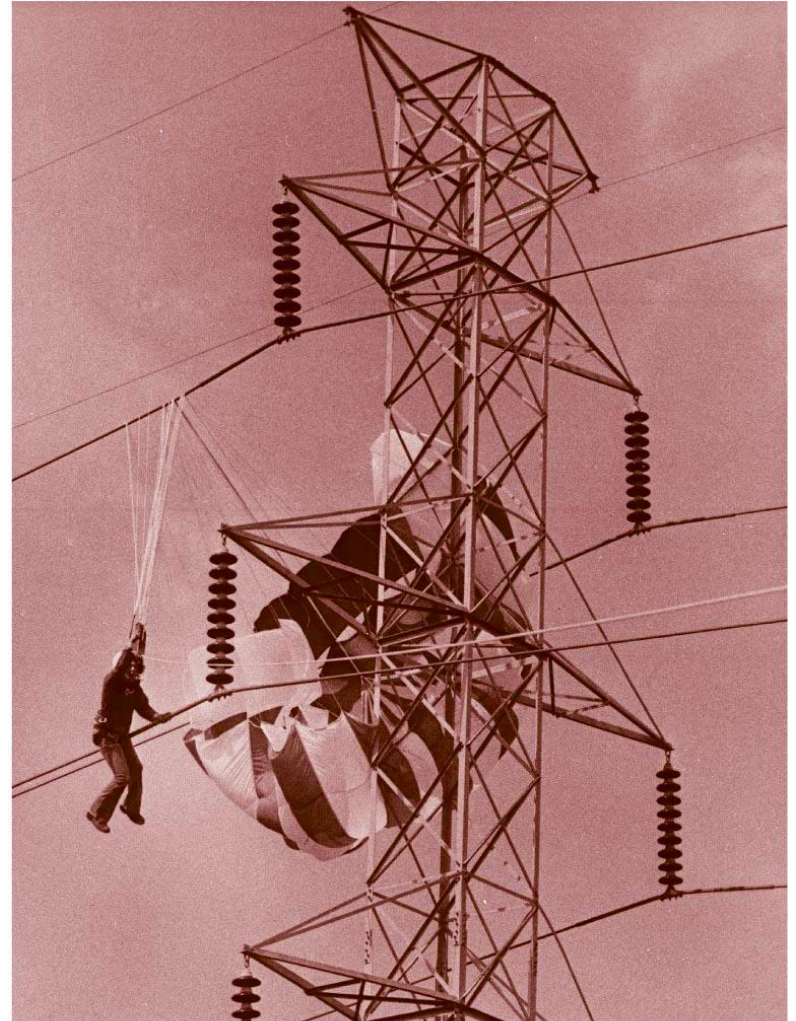


Ch24 Electric potential (电位)

- All about energy associated with electrical interactions

This parasailer (滑翔伞) landed on a 138,000-volt (V, 伏) power line. Why didn't he get electrocuted?

He touches only 1 line – there's no potential differences & hence no energy transfer involved.



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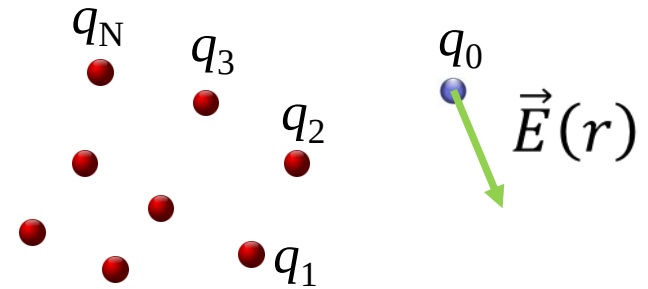
Today's topics (Chapter 24)

- Electric Potential Energy
- Electric Potential
- Various Models
- Calculating Electric Field by Potential

Motion of a Charged Particle

How to obtain the motion of q ?

$$\vec{F} = q\vec{E}(r) = m \frac{d^2\vec{r}}{dt^2} = m \frac{d\vec{v}}{dt}$$



In principle, you can always apply the above formula and get the trajectory.

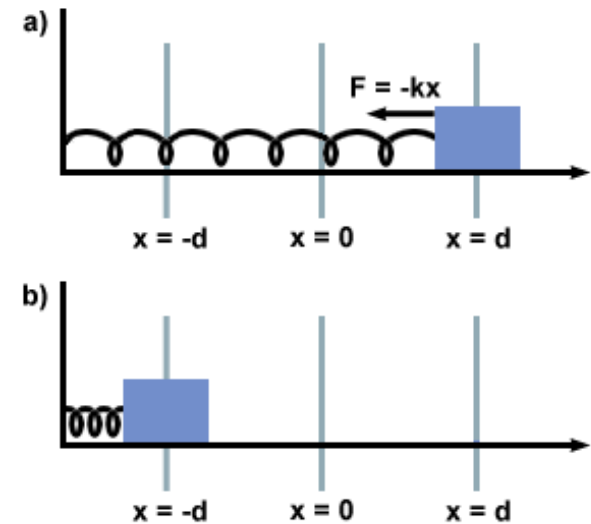
Mostly, we just want to know the initial and final states, do not care about the moving course.

Recap: Work-Energy Theorem

Newton's Law (scalar form)

$$m \frac{dv}{dt} = F \Rightarrow v m \frac{dv}{dt} = F v$$

$$\Rightarrow \frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = F \frac{dx}{dt}$$



Integral on both sides: **Work-Energy Theorem**

$$\frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 = \int_{x_1}^{x_2} F(x) dx$$

Work-Energy Theorem is true for all forces!

Recap: Conservation of Energy

$$K_2 - K_1 = \int_{x_1}^{x_2} F(x) dx = G(x_2) - G(x_1), \quad \frac{dG}{dx} = F$$

$$\Rightarrow K_2 - G(x_2) = K_1 - G(x_1), \quad \text{Let } U(x) = -G(x)$$

One gets conservation of energy

$$K_2 + U(x_2) = K_1 + U(x_1), \quad F = -\frac{dU}{dx}$$

$$U(x_2) - U(x_1) = -\int_{x_1}^{x_2} F(x) dx$$

Why the above derivation does NOT hold for friction?

Definition

Coulomb's law: $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r}$

- **Electric Field ($\vec{E}$)**

- Force per unit charge at a point in space
- a vector field

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

- **Electric potential (V)**

- Electrostatic potential energy U per unit charge at a point in space
- a scalar field

$$V = \frac{U}{q_0}$$

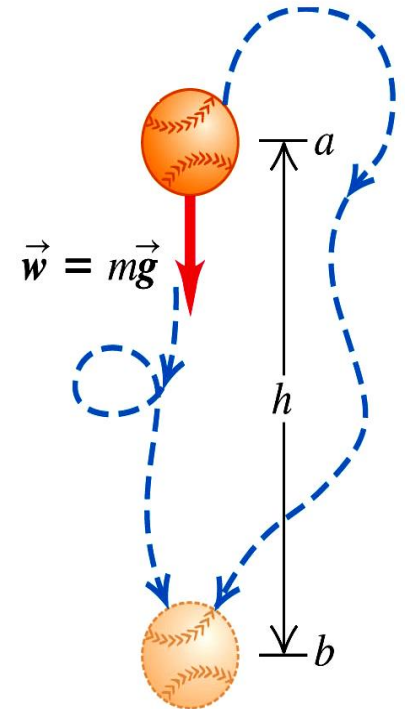
Energy : work done (W) by a force

$$W_{a \rightarrow b} = \int_a^b \vec{F} \cdot d\vec{l} = \int_a^b F \cos \phi \, dl \quad (\text{work done by a force})$$

Work done $\longleftrightarrow$ Potential energy

$$W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U$$

(work done by a conservative force)



Work, Energy and their Relations

Work done by the electric field:

$$W = -\Delta U = -q(V_f - V_i)$$

Conservation of energy:

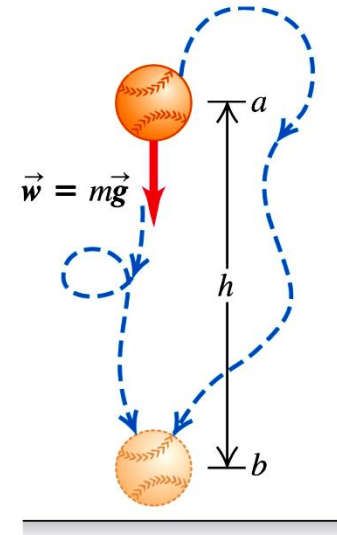
$$U_i + K_i = U_f + K_f$$

Work by an external force W_{app} :

$$U_i + K_i + W_{\text{app}} = U_f + K_f$$

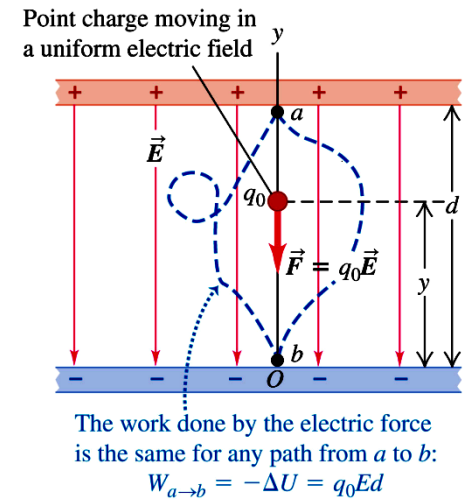
Definition – conservation force (守恒力)

1. A conservative force does zero total work on any closed path
2. The work done in moving a particle between two points is independent of the path taken



Example of non-conservation force?

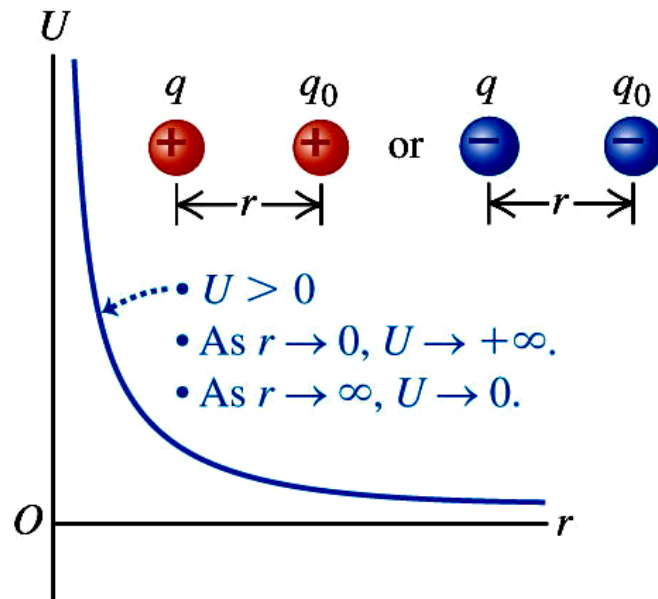
Friction, air drag



Definition - work done on q_0 by q

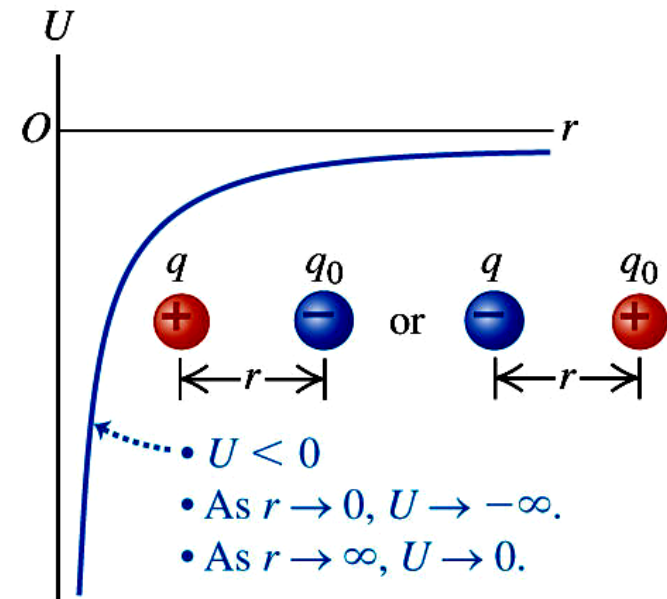
- We can always assign an electric potential energy U when electrostatic force acts between two or more charge particles
- Potential energy is always defined relative to some reference point where $U = 0$

(a) q and q_0 have the same sign.



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(b) q and q_0 have opposite signs.



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- Potential energy can be only defined for the whole system, not for any individual charge.

Today's topics (Chapter 24)

- Electric Potential Energy
- Electric Potential
- Various Models
- Calculating Electric Field by Potential

Definition

(work done by a
conservative force)

$$W_{a \rightarrow b} = U_a - U_b = -(U_b - U_a) = -\Delta U$$

We can define the electric potential energy U at point a in the following way:

work done by the E-field as the charged particle moves in from infinity ∞ to point a

$$U = -W_{\infty}$$

→ Electric potential
(i.e., potential energy
per unit charge)

$$V = -\frac{W_{\infty}}{q}$$

Definition – potential difference in a field

Potential difference (V) between point i and point f : $V_f - V_i$?

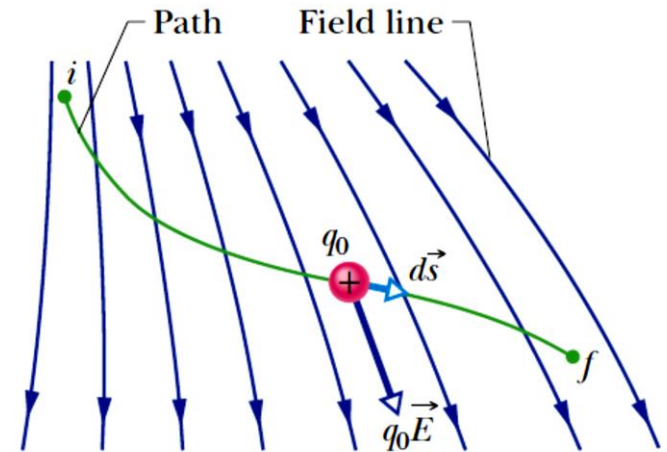
Potential energy difference (U) between point i and point f : $\Delta U = U_f - U_i = -W_{\text{by the field}}$

$$dW = \vec{F} \cdot d\vec{s}$$

$$dW = q_0 \vec{E} \cdot d\vec{s}$$

$$W = q_0 \int_i^f \vec{E} \cdot d\vec{s}$$

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$



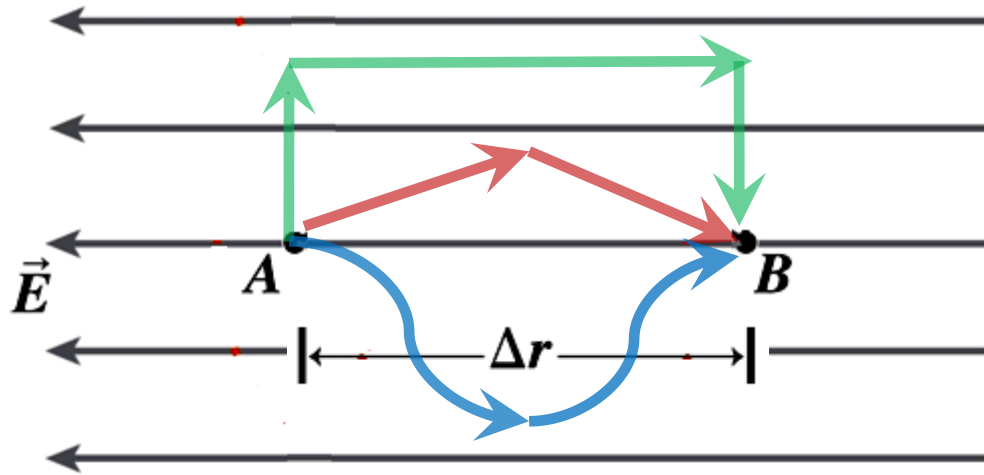
If i is at ∞ where U is 0, V_i is 0.

$$V = - \int_i^f \vec{E} \cdot d\vec{s}$$

Potential does not depend on trajectory

Potential Difference is Path Independent

Potential difference ΔV_{AB} depends only on positions of A & B .



Calculating along any paths (1, 2, or 3) gives $\Delta V_{AB} = E \Delta r$.

Electric potential of a point charge

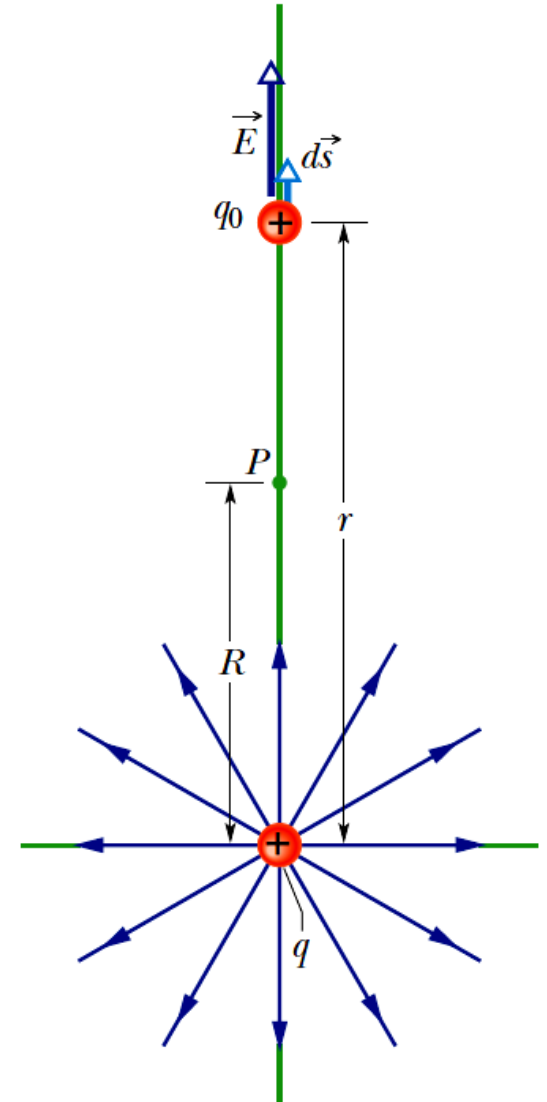
$$V_f - V_i = - \int_{\infty}^R E dr$$

as $V_i = 0$ at ∞ and $E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

$$V = - \frac{q}{4\pi\epsilon_0} \int_{\infty}^R \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{\infty}^R$$

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

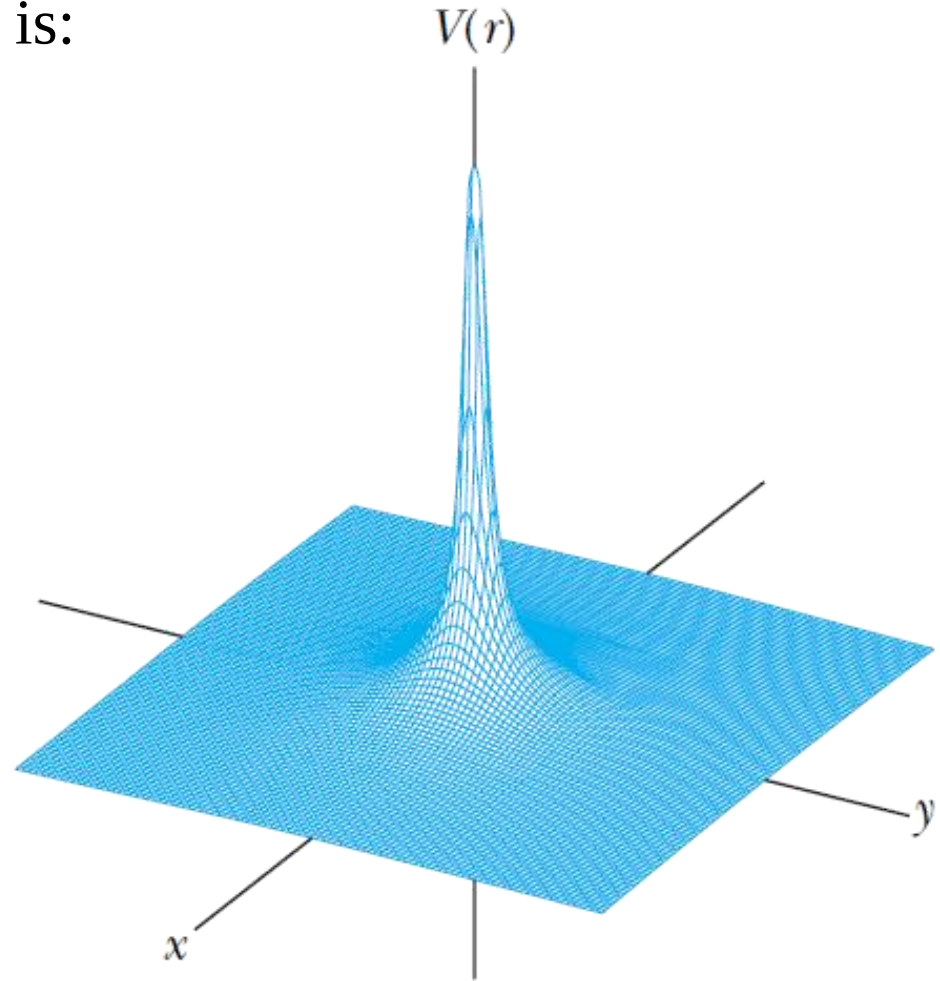
How about the V of electric dipole???



Electric potential of a point charge

So the potential due to a point charge q at distance R is:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$$



Force, Field, Potential Energy & Electric Potential

Quantity	Symbol	Units
Force	$\mathbf{F}$	N
Electric field	$\mathbf{E} = \mathbf{F} / q$	N/C or V/m
Potential energy difference	$\Delta U = -\int_A^B \mathbf{F} \cdot d\mathbf{r}$	J
Electric potential difference	$\Delta V = \frac{\Delta U}{q}$ $\Delta V = -\int_A^B \mathbf{E} \cdot d\mathbf{r}$	J/C or V

Example

$$\Delta V = -\int_A^B \mathbf{E} \cdot d\mathbf{r}$$

What would happen to V_{AB} in the figure if

doubles

(a) E were doubled;

doubles

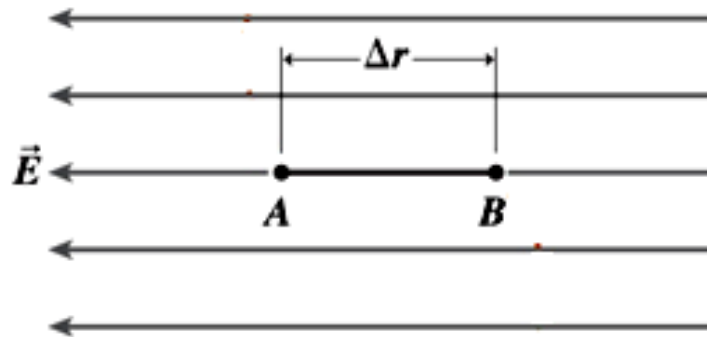
(b) Δr were doubled;

becomes 0

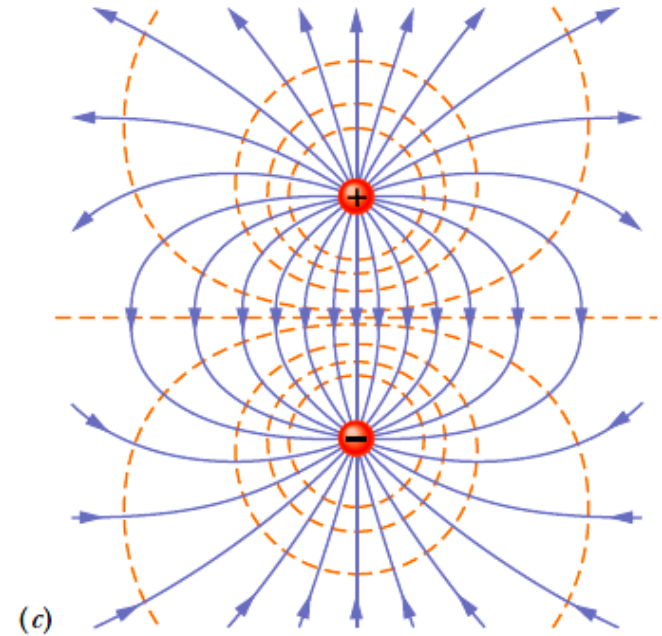
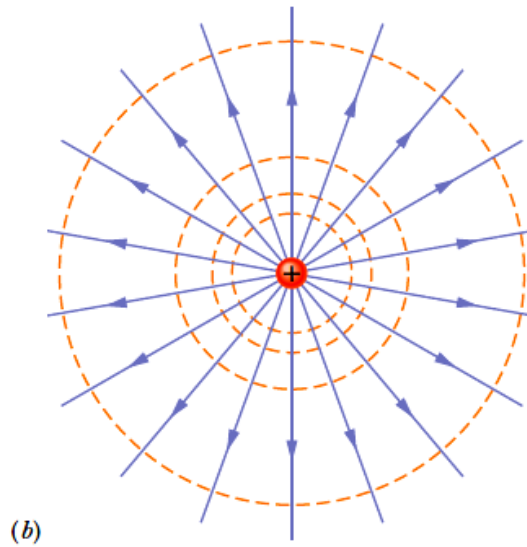
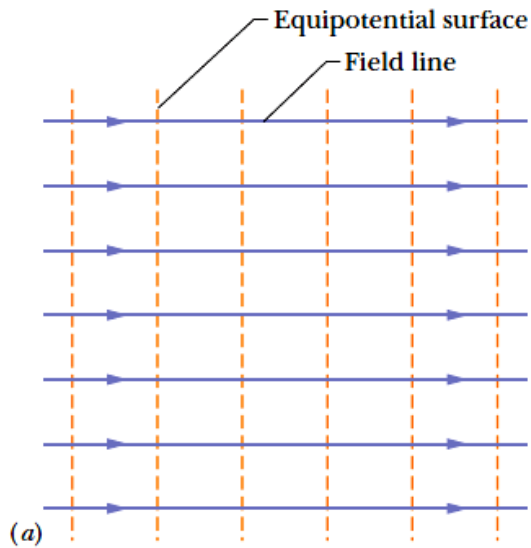
(c) the points were moved so the path lay at right angles to $\mathbf{E}$;

reverses sign

(d) the positions of A & B are interchanged.

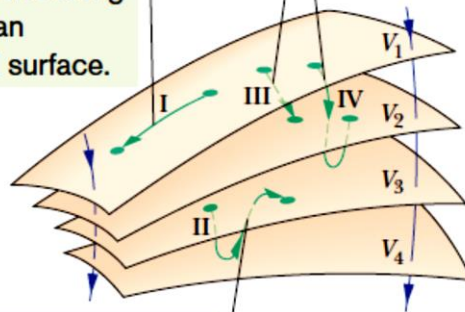


Equipotential surface (等势面)



No work is done along this path on an equipotential surface.

Equal work is done along these paths between the same surfaces.



No work is done along this path that returns to the same surface.

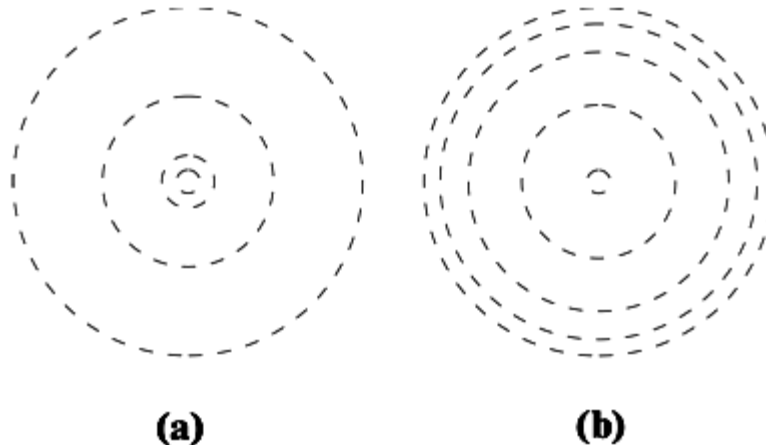
E must be $\perp$ to the equipotential surface

Example

The figure shows cross sections through 2 equipotential surfaces.

In both diagrams, the potential difference between adjacent equipotentials is the same.

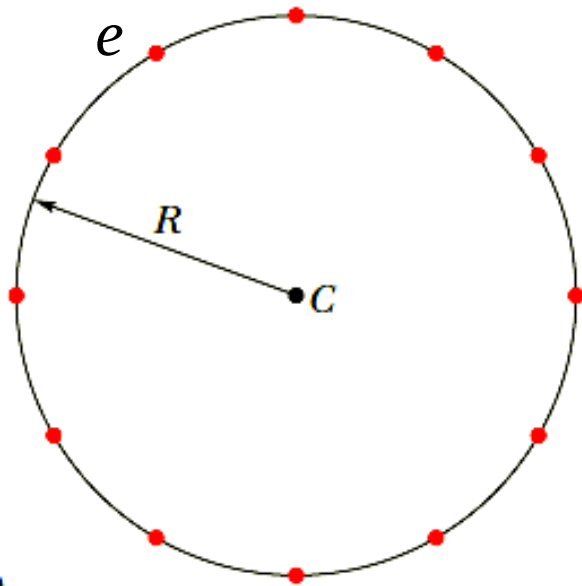
Which could represent the field of a point charge? Explain.



(a). Potential decreases as r^{-1} , so the spacings between equipotentials should increase with r .

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Example



The electric potential V at C ?

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i} \quad (n \text{ point charges})$$

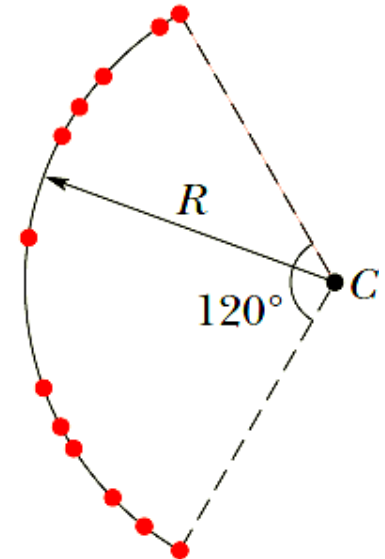
$$V = -12 \frac{1}{4\pi\epsilon_0} \frac{e}{R}$$

What if ?

How about E -field at C ?

$$\vec{E} = 0$$

All cancel out by the
opposite charge !



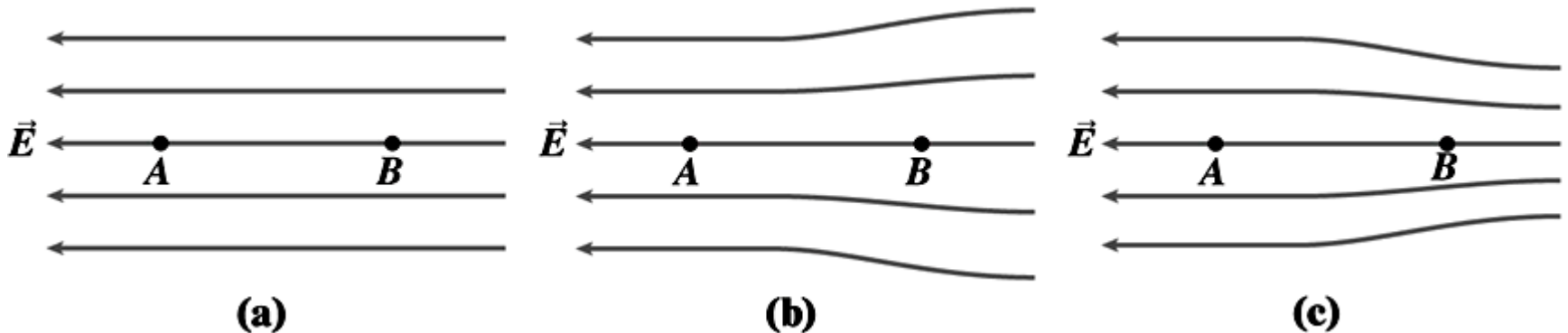
Example

The figure shows three straight paths AB of the same length, each in a different electric field.

The field at A is the same in each.

Rank the potential differences ΔV_{AB} .

$$\Delta V_{AB} \equiv \frac{\Delta U_{AB}}{q} \quad \Delta U_{AB} = U_B - U_A$$



Smallest ΔV_{AB} .

Largest ΔV_{AB} .

The Volt & the Electronvolt

$$[V] = J/C = \text{Volt} = V$$

$$\Delta U = q \Delta V$$

E.g., for a 12V battery, 12J of work is done on every 1C charge that moves from its negative to its positive terminals.

Electronvolt (eV) = **energy** gained by a particle carrying 1 elementary charge when it moves through a potential difference of 1 volt.

$$1 \text{ elementary charge} = 1.6 \times 10^{-19} \text{ C} = e$$

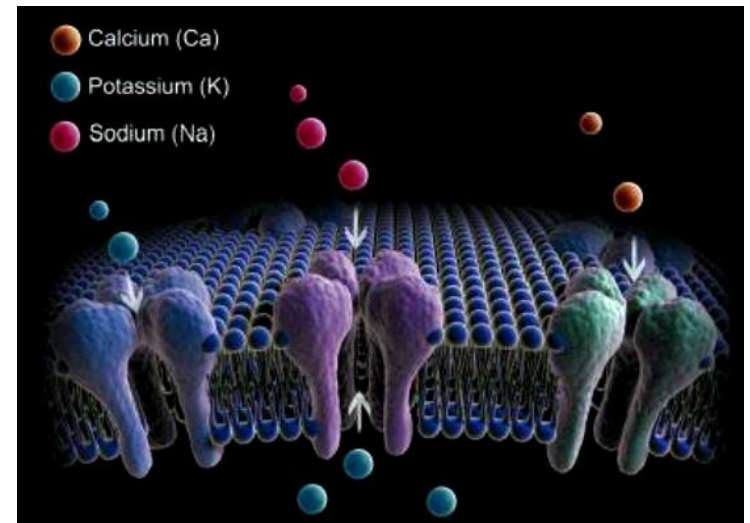
$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

International Voltage

Afghanistan	220 V
Australia	240 V
Bahamas	120 V
Brazil	120/220 V
Canada	120 V
China	220 V
Finland	230 V
Guam	110 V
Japan	100 V
Mexico	127 V
Spain	230 V
United Kingdom	230 V
United States	120 V
Vietnam	127/220 V

Typical Potential Differences

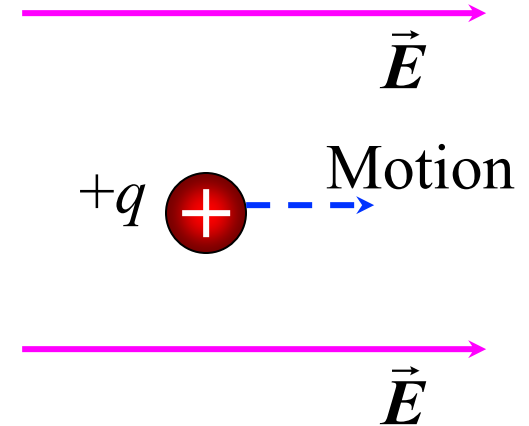
Between human arm & leg due to heart's electrical activity	1 mV
Across biological cell membrane	80 mV
Between terminals of flashlight battery	1.5 V
Car battery	12 V
Electric outlet (depends on country)	100-240 V
Between electric transmission line & ground	365 kV
Between base of thunderstorm cloud & ground	100 MV





When a positive charge moves in the direction of the electric field,

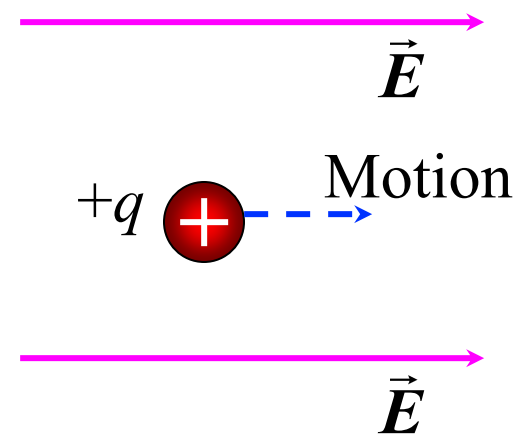
- A. the field does positive work on it and the potential energy increases.
- B. the field does positive work on it and the potential energy decreases.
- C. the field does negative work on it and the potential energy increases.
- D. the field does negative work on it and the potential energy decreases.



Q1 Answer

When a positive charge moves in the direction of the electric field,

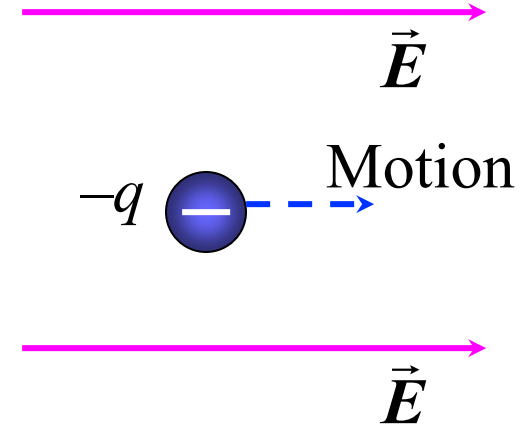
- A. the field does positive work on it and the potential energy increases.
- ✓ B. the field does positive work on it and the potential energy decreases.
- C. the field does negative work on it and the potential energy increases.
- D. the field does negative work on it and the potential energy decreases.





When a negative charge moves in the direction of the electric field,

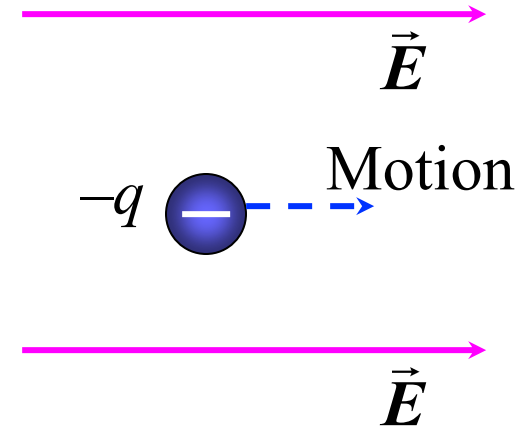
- A. the field does positive work on it and the potential energy increases.
- B. the field does positive work on it and the potential energy decreases.
- C. the field does negative work on it and the potential energy increases.
- D. the field does negative work on it and the potential energy decreases.



Q2 answer

When a negative charge moves in the direction of the electric field,

- A. the field does positive work on it and the potential energy increases.
- B. the field does positive work on it and the potential energy decreases.
- ✓ C. the field does negative work on it and the potential energy increases.
- D. the field does negative work on it and the potential energy decreases.

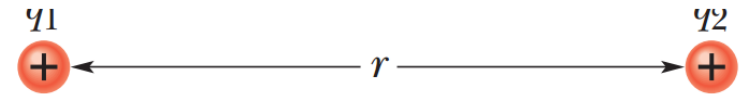


Today's topics (Chapter 24)

- Electric Potential Energy
- Electric Potential
- Various Models
- Calculating Electric Field by Potential

Model I: Two-Particle System

Fix q_2 and move q_1 from $r = \infty$ to its final place



$$U_f - U_i = q_1 (V_f - V_i), \quad \text{where } V_i = 0, V_f = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r}$$

$$U = U_f - U_i = \frac{q_1 q_2}{4\pi\epsilon_0 r}$$

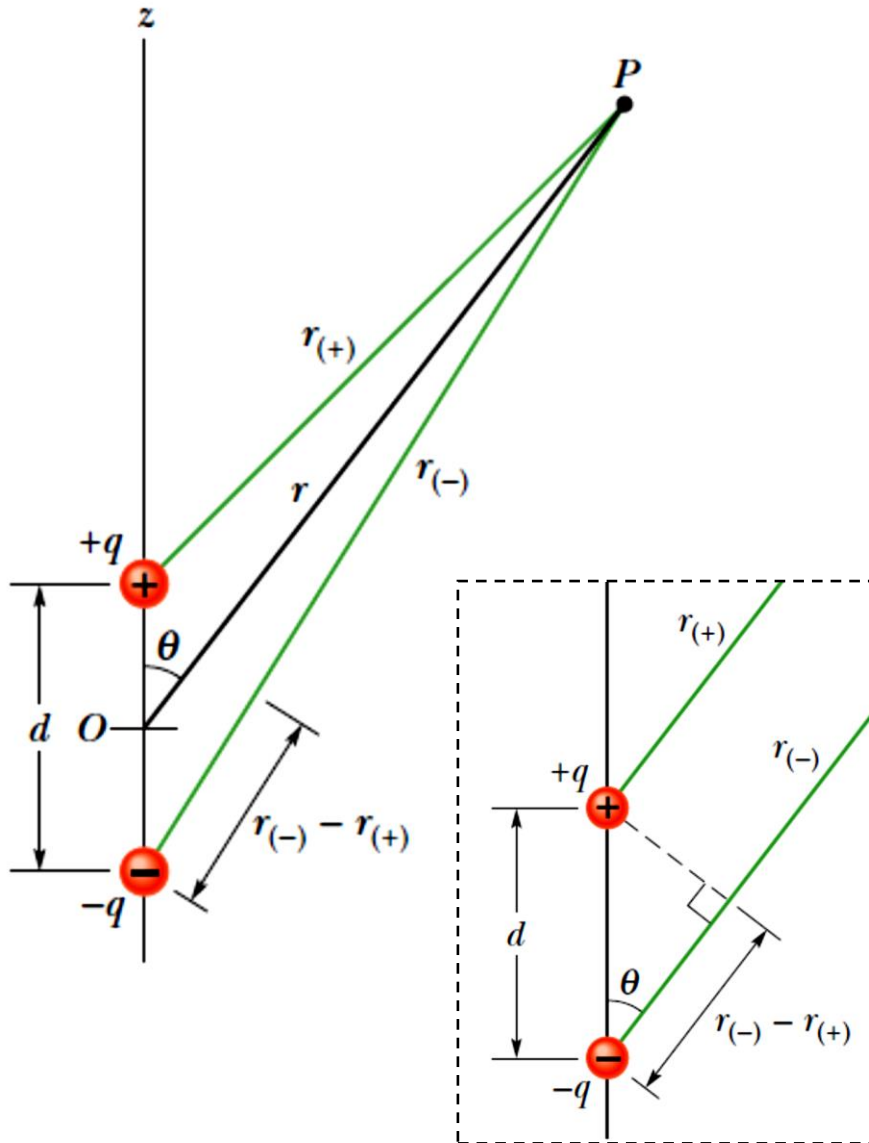
Includes signs of q

For some historical reasons, sometimes we use U to represent the potential energy difference, in particular when referenced to infinity. **But you can always use ΔU to avoid confusion.**

If q_1 and q_2 are not positive, what is the potential energy?

It is still that form!

Potential due to electric dipole



$$\begin{aligned} V &= \sum_{i=1}^2 V_i = V_{(+)} + V_{(-)} \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}} \right) \\ &= \frac{q}{4\pi\epsilon_0} \frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}} \end{aligned}$$

$$V = \frac{q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2}$$

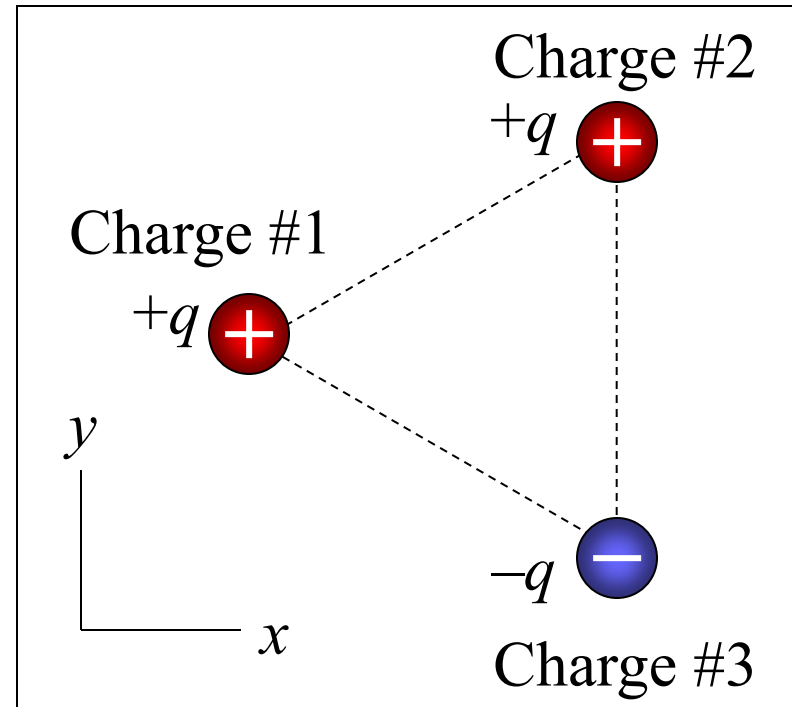
$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

p is the magnitude of the electric dipole moment $\vec{p}$
($p = qd$)



The electric potential energy of two point charges approaches zero as the two point charges move farther away from each other.

If the three point charges shown here lie at the vertices of an equilateral triangle, the electric potential energy of the system of three charges is



A. positive.

B. negative.

C. zero.

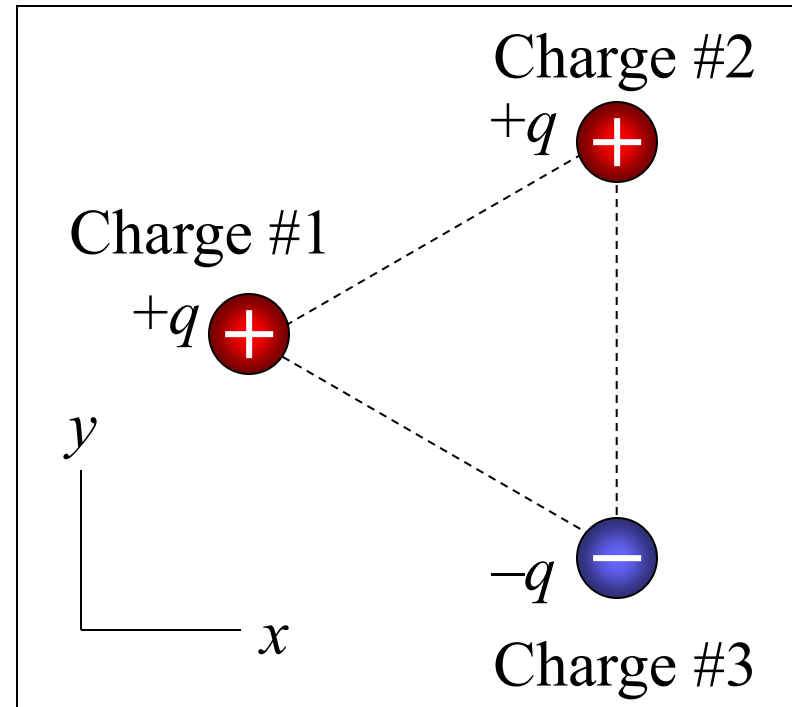
D. not enough information given to decide

$$W_{a \rightarrow b} = U_a - U_b$$

Q3 answer

The electric potential energy of two point charges approaches zero as the two point charges move farther away from each other.

If the three point charges shown here lie at the vertices of an equilateral triangle, the electric potential energy of the system of three charges is



- A. positive.
- ✓ B. negative.
- C. zero.
- D. not enough information given to decide

Famous Bohr Model

In the Bohr model of a hydrogen atom, the electron, if it is in the ground state, orbits the proton at a distance of $r = 5.29 \times 10^{-11} \text{ m}$. Find the ionization energy, i.e. the energy required to remove the electron from the atom.

Given:

$$\begin{aligned} r &= 5.292 \times 10^{-11} \text{ m} \\ m_e &= 9.11 \times 10^{-31} \text{ kg} \\ m_p &= 1.67 \times 10^{-27} \text{ kg} \\ |e| &= 1.60 \times 10^{-19} \text{ C} \end{aligned}$$

Find:

$$E = ?$$

The ionization energy equals to the total energy of the electron-proton system,

$$E = U + K \text{ with } U = -k \frac{e^2}{r}, K = \frac{mv^2}{2}$$

The velocity of e can be found by analyzing the force on the electron. This force is the Coulomb force; because the electron travels in a circular orbit, the acceleration will be the centripetal acceleration:

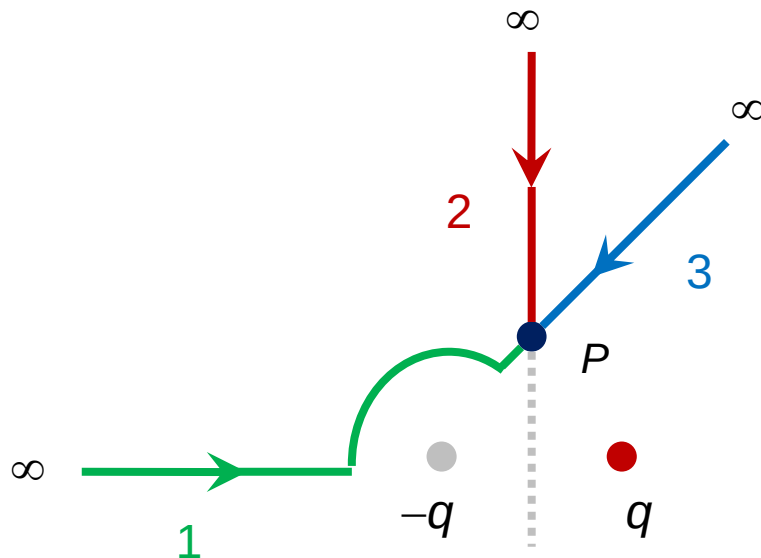
$$ma_c = F_c \Rightarrow \frac{mv^2}{r} = k \frac{e^2}{r^2} \Rightarrow v^2 = k \frac{e^2}{mr}$$

$$E = -k \frac{e^2}{r} + \frac{m}{2} \left(k \frac{e^2}{mr} \right) = -k \frac{e^2}{2r} = -2.18 \times 10^{-18} \text{ J} \approx -13.6 \text{ eV}$$

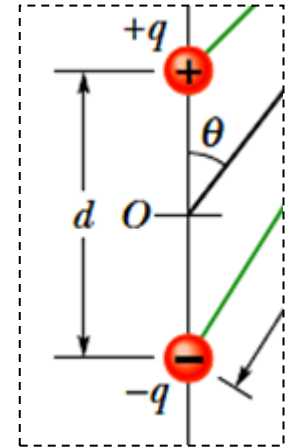
Example

The figure shows 3 paths from infinity to a point P on a dipole's perpendicular bisector.

Compare the work done in moving a charge to P on each of the paths.



$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$



V is path independent

→ work on all 3 paths are the same.

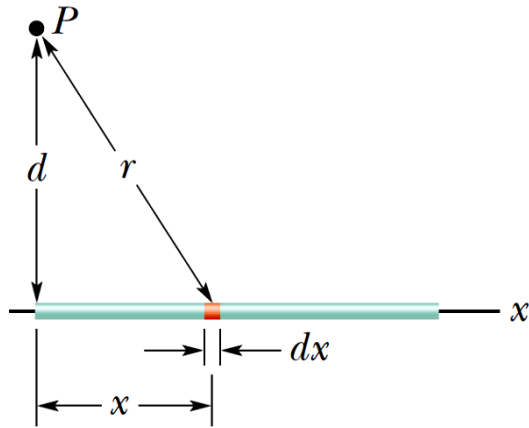
Work along path 2 is 0 since $V = 0$ on it.

Hence, $W = 0$ for all 3 paths.

Potential Due to a Continuous Charge Distribution

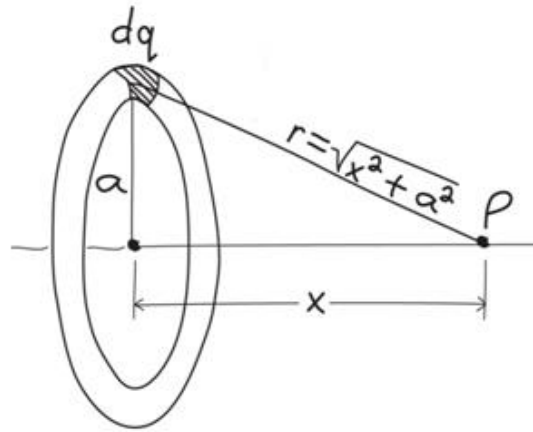
$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r}$$

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$



$$dV = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{(x^2 + d^2)^{1/2}}$$

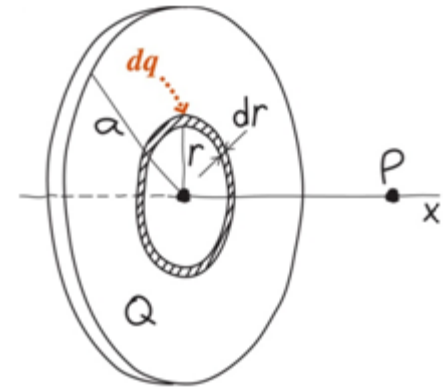
λ is the linear charge density



Same r for all dq

$$V(x) = \int k \frac{dq}{r}$$

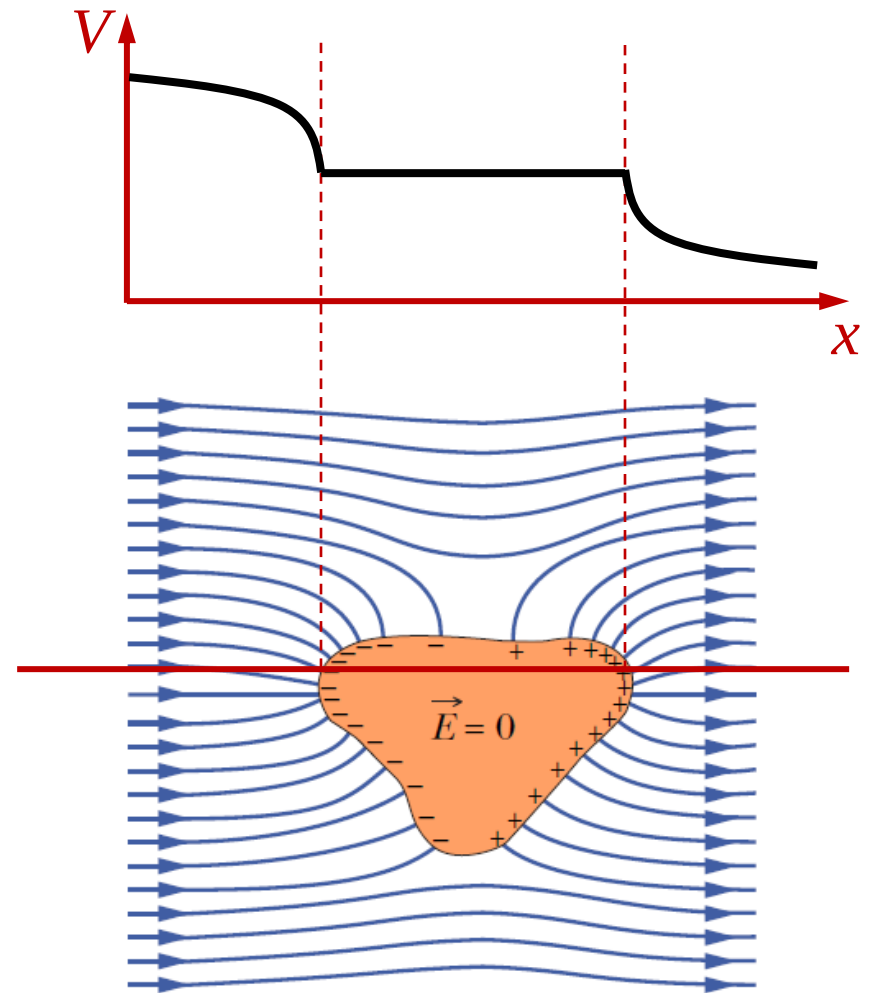
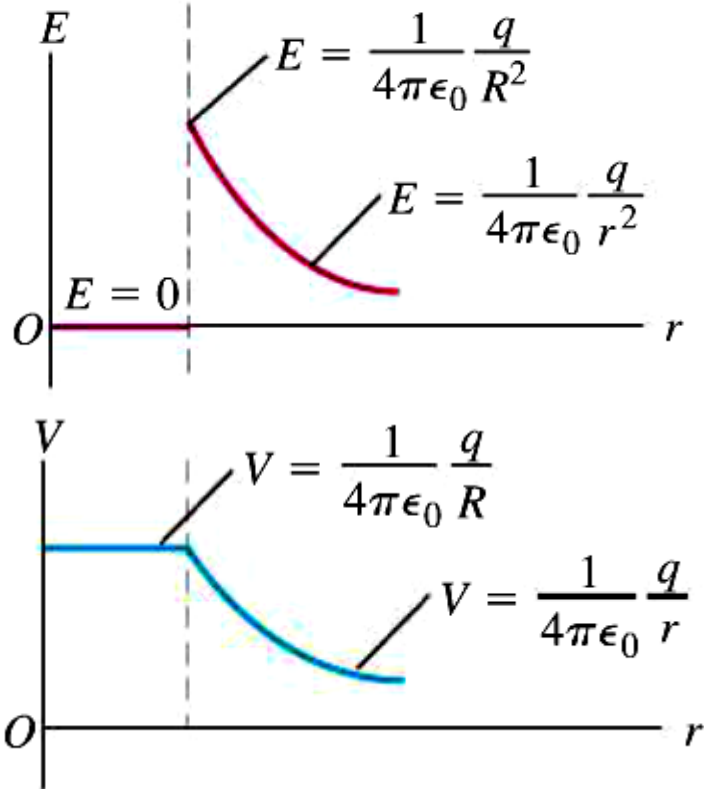
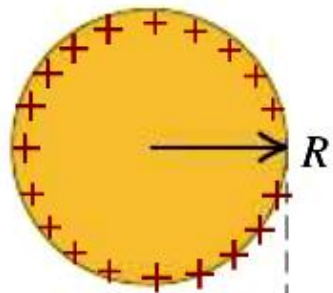
$$= \frac{k}{\sqrt{x^2 + a^2}} \int dq$$



$$V(x) = \int dV = \int \frac{k}{\sqrt{x^2 + r^2}} dq$$

$$= \int_0^a \frac{k}{\sqrt{x^2 + r^2}} \left(\frac{Q}{\pi a^2} \right) 2\pi r dr$$

Potential of a Charged Isolated Conductor



Today's topics (Chapter 24)

- Electric Potential Energy
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- Calculating Electric Field by Potential

Calculating the Field from the Potential

$$V \rightarrow \vec{E}$$

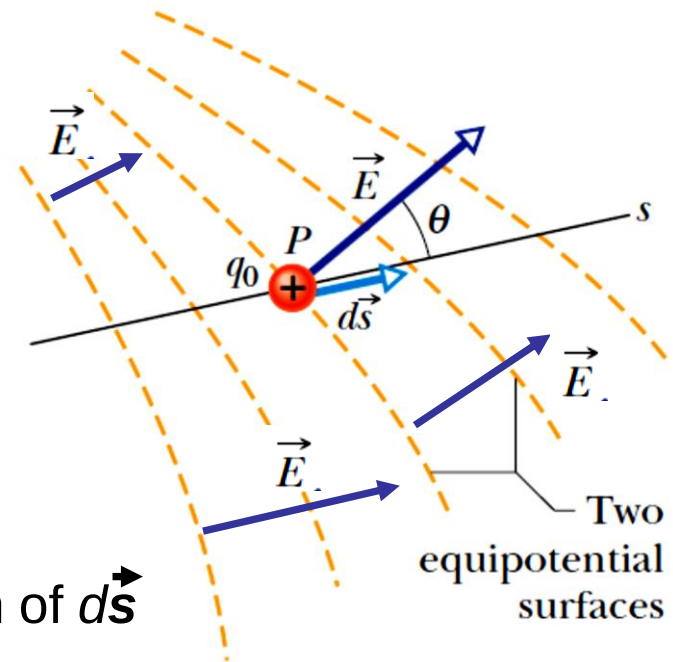
$$V = \frac{U}{q_0} = - \int_i^f \vec{E} \cdot d\vec{s}$$

$$q_0 dV = - q_0 E(\cos \theta) ds$$

$$E \cos \theta = - \frac{dV}{ds}$$

$$E_s = - \frac{\partial V}{\partial s}$$

E_s in the direction of $d\vec{s}$



$\vec{E}$ is the negative rate of change of potential

$$E_x = - \frac{\partial V}{\partial x} \quad E_y = - \frac{\partial V}{\partial y} \quad E_z = - \frac{\partial V}{\partial z} \quad (\text{components of } \vec{E} \text{ in terms of } V)$$

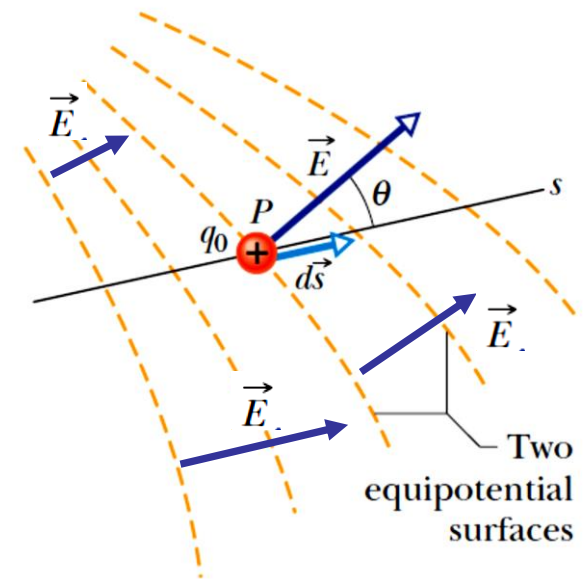
Calculating the Field from the Potential

$$V \rightarrow \vec{E}$$

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

$$\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right)$$

($\vec{E}$ in terms of V)



Gradient of the function f

$$\vec{\nabla}f = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right)f$$

$\vec{\nabla}$ is called “grad” or “del”

$$\vec{E} = -\vec{\nabla}V$$

$\vec{E}$ is the –negative gradient of V

Example for $V \rightarrow \vec{E}$

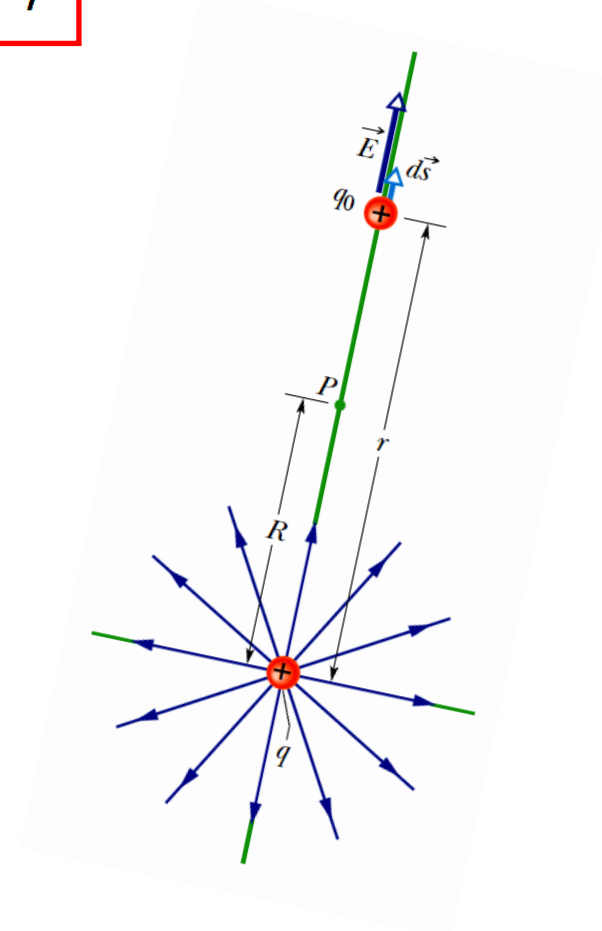
$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$E_r = -\frac{\partial V}{\partial r}$$

$$= -\frac{\partial}{\partial r} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\vec{E} = \hat{r}E_r = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

What if we use x, y, z ?



Example for $V \rightarrow \vec{E}$ in x, y, z

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

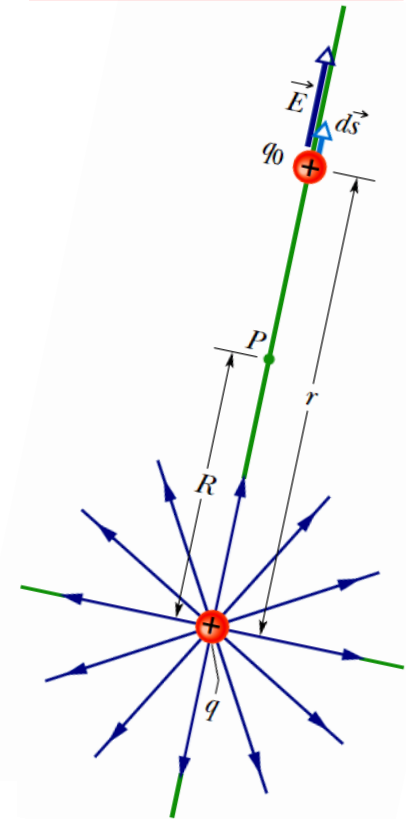
$$\begin{aligned} \frac{\partial V}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\sqrt{x^2 + y^2 + z^2}} \right) \\ &= -\frac{1}{4\pi\epsilon_0} \frac{qx}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{qx}{4\pi\epsilon_0 r^3} \end{aligned}$$

Similarly

$$\frac{\partial V}{\partial y} = -\frac{qy}{4\pi\epsilon_0 r^3} \quad \frac{\partial V}{\partial z} = -\frac{qz}{4\pi\epsilon_0 r^3}$$

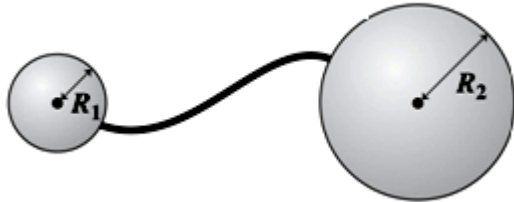
By : $\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right)$

$$\begin{aligned} \vec{E} &= -\left[\hat{i}\left(-\frac{qx}{4\pi\epsilon_0 r^3}\right) + \hat{j}\left(-\frac{qy}{4\pi\epsilon_0 r^3}\right) + \hat{k}\left(-\frac{qz}{4\pi\epsilon_0 r^3}\right)\right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \left(\frac{x\hat{i} + y\hat{j} + z\hat{k}}{r}\right) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \end{aligned}$$



Example

Consider 2 widely separated, charged conducting spheres.



Their potentials are $V_1 = k \frac{Q_1}{R_1}$ $V_2 = k \frac{Q_2}{R_2}$

What will happen if we connect them with a thin conducting wire?

There'll be charge transfer until $V_1 = V_2$, i.e., $\frac{Q'_1}{R_1} = \frac{Q'_2}{R_2}$

In terms of the surface charge densities $\sigma_j = \frac{Q'_j}{4\pi R_j^2}$

we have $\sigma_1 R_1 = \sigma_2 R_2 \rightarrow E_{1\perp} R_1 = E_{2\perp} R_2$

Smaller sphere has higher field at surface.

An Irregular Conductor

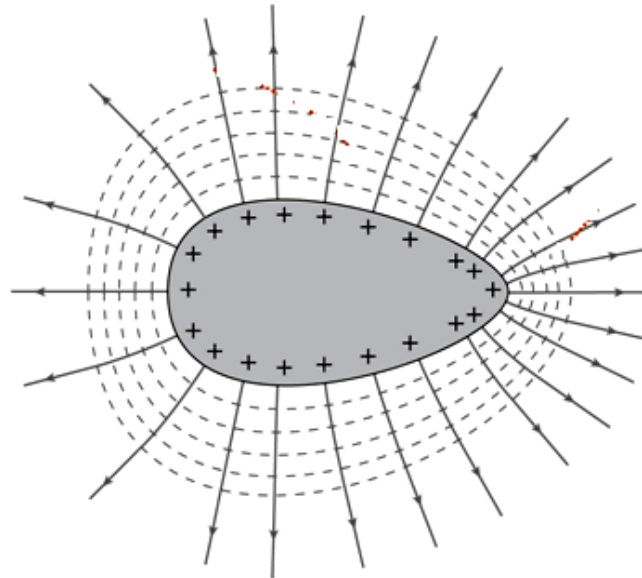
Sketch some equipotentials & electric field lines for an isolated egg-shaped conductor.

Ans.

Surface is equipotential $\rightarrow |E_{\perp}|$ is larger where curvature of surface is large. **Why?**

More field lines emerging from sharply curved regions.

From afar, conductor is like a point charge.

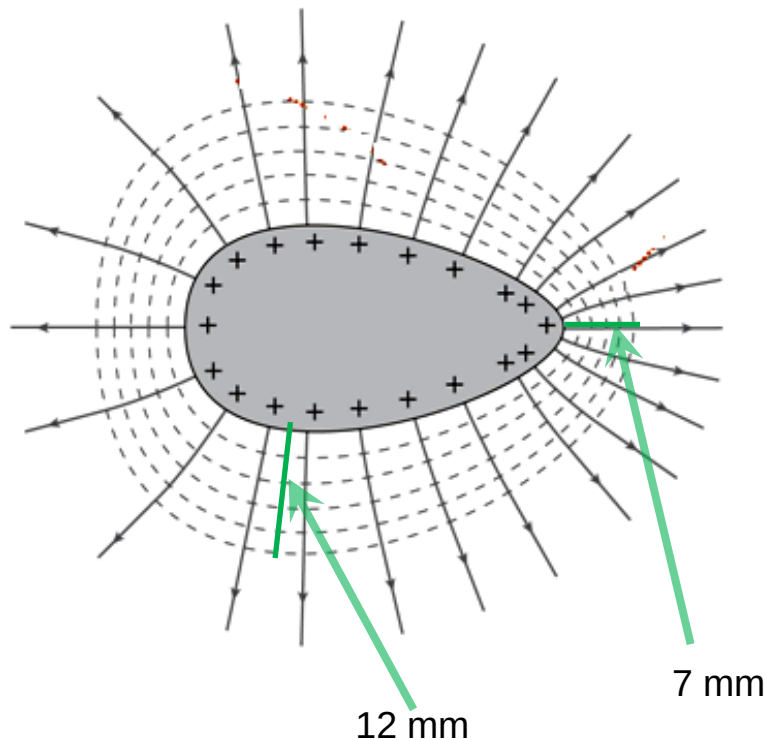


Making the Connection

$$E = V/m$$

The potential difference between the conductor and the outermost equipotential shown in figure is 70 V.

Determine approximate values for the strongest & weakest electric fields in the region, assuming it is drawn at the sizes shown.



Strongest field :

$$E = \frac{70 \text{ V}}{7 \times 10^{-3} \text{ m}} = 10 \text{ kV / m}$$

Weakest field :

$$E = \frac{70 \text{ V}}{12 \times 10^{-3} \text{ m}} = 5.8 \text{ kV / m}$$

Q4

A hollow metal sphere is charged to a potential V .
The potential at its center is:

A) V

B) 0

C) $-V$

D) $2V$

E) pV

Summary

Review & Summary

Electric Potential The electric potential V at a point P in the electric field of a charged object is

$$V = \frac{-W_{\infty}}{q_0} = \frac{U}{q_0}, \quad (24-2)$$

where W_{∞} is the work that would be done by the electric force on a positive test charge were it brought from an infinite distance to P , and U is the potential energy that would then be stored in the test charge-object system.

Electric Potential Energy If a particle with charge q is placed at a point where the electric potential of a charged object is V , the electric potential energy U of the particle-object system is

$$U = qV. \quad (24-3)$$

If the particle moves through a potential difference ΔV , the change in the electric potential energy is

$$\Delta U = q \Delta V = q(V_f - V_i). \quad (24-4)$$

Mechanical Energy If a particle moves through a change ΔV in electric potential without an applied force acting on it, applying the conservation of mechanical energy gives the change in kinetic energy as

$$\Delta K = -q \Delta V. \quad (24-9)$$

If, instead, an applied force acts on the particle, doing work W_{app} , the change in kinetic energy is

$$\Delta K = -q \Delta V + W_{\text{app}}. \quad (24-11)$$

In the special case when $\Delta K = 0$, the work of an applied force

Summary

involves only the motion of the particle through a potential difference:

$$W_{\text{app}} = q \Delta V \quad (\text{for } K_i = K_f). \quad (24-12)$$

Equipotential Surfaces The points on an **equipotential surface** all have the same electric potential. The work done on a test charge in moving it from one such surface to another is independent of the locations of the initial and final points on these surfaces and of the path that joins the points. The electric field $\vec{E}$ is always directed perpendicularly to corresponding equipotential surfaces.

Finding V from $\vec{E}$ The electric potential difference between two points i and f is

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}, \quad (24-18)$$

where the integral is taken over any path connecting the points. If the integration is difficult along any particular path, we can choose a different path along which the integration might be easier. If we choose $V_i = 0$, we have, for the potential at a particular point,

$$V = - \int_i^f \vec{E} \cdot d\vec{s}. \quad (24-19)$$

In the special case of a uniform field of magnitude E , the potential change between two adjacent (parallel) equipotential lines separated by distance Δx is

$$\Delta V = -E \Delta x. \quad (24-21)$$

Potential Due to a Charged Particle The electric potential due to a single charged particle at a distance r from that particle is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}, \quad (24-26)$$

where V has the same sign as q . The potential due to a collection of charged particles is

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}. \quad (24-27)$$

Potential Due to an Electric Dipole At a distance r from an electric dipole with dipole moment magnitude $p = qd$, the electric potential of the dipole is

$$V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2} \quad (24-30)$$

for $r \gg d$; the angle θ is defined in Fig. 24-13.

Potential Due to a Continuous Charge Distribution

For a continuous distribution of charge, Eq. 24-27 becomes

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}, \quad (24-32)$$

in which the integral is taken over the entire distribution.

Calculating $\vec{E}$ from V The component of $\vec{E}$ in any direction is the negative of the rate at which the potential changes with distance in that direction:

$$E_s = - \frac{\partial V}{\partial s}. \quad (24-40)$$

The x , y , and z components of $\vec{E}$ may be found from

$$E_x = - \frac{\partial V}{\partial x}; \quad E_y = - \frac{\partial V}{\partial y}; \quad E_z = - \frac{\partial V}{\partial z}. \quad (24-41)$$

When $\vec{E}$ is uniform, Eq. 24-40 reduces to

$$E = - \frac{\Delta V}{\Delta s}, \quad (24-42)$$

where s is perpendicular to the equipotential surfaces.

Electric Potential Energy of a System of Charged Particles

The electric potential energy of a system of charged particles is equal to the work needed to assemble the system with the particles initially at rest and infinitely distant from each other. For two particles at separation r ,

$$U = W = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}. \quad (24-46)$$

Potential of a Charged Conductor An excess charge placed on a conductor will, in the equilibrium state, be located entirely on the outer surface of the conductor. The charge will distribute itself so that the following occur: (1) The entire conductor, including interior points, is at a uniform potential. (2) At every internal point, the electric field due to the charge cancels the external electric field that otherwise would have been there. (3) The net electric field at every point on the surface is perpendicular to the surface.